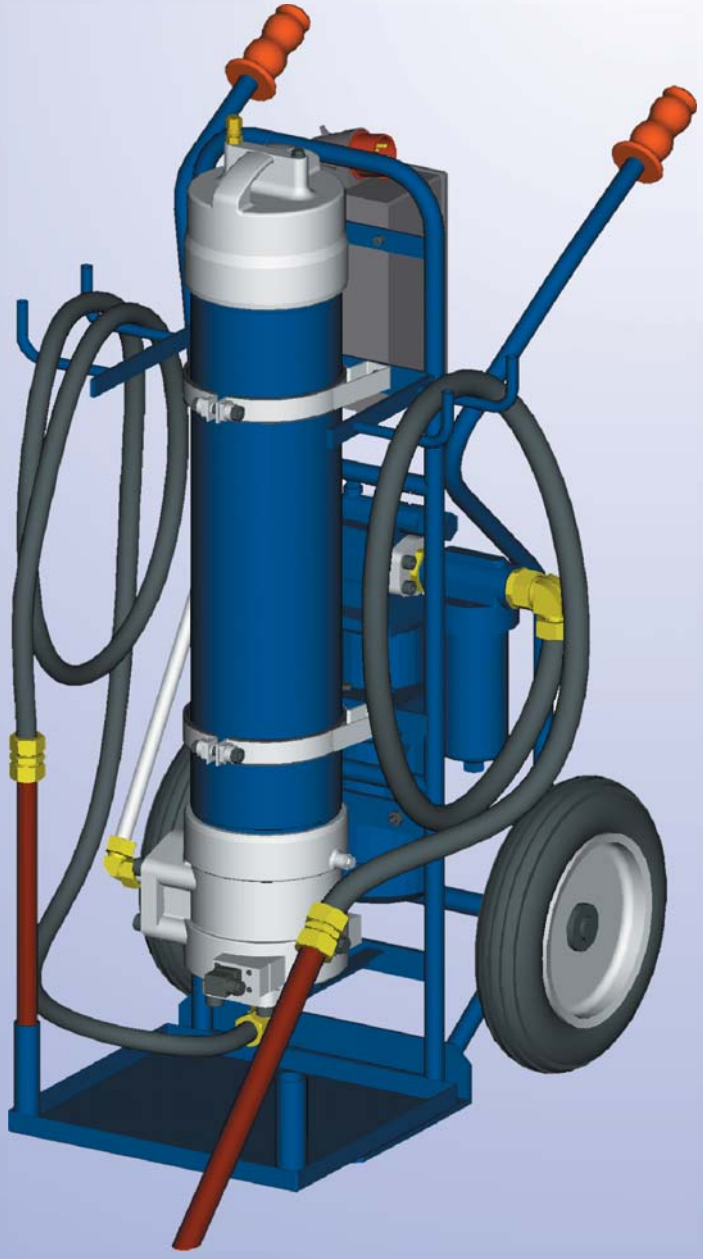
EPE

Filters . Accumulators

Offline Filtration Systems

15...170 NFF2 0020-0270



Mobile filtration systems with integral motor-pump units

For temporary or permanent installation using offline flow

Reduce usage of main flow filters, extending life of more expensive elements

Filtration of fresh and top-up fluids.

Flushing of contaminated systems or reservoirs

Upgrade filtration on existing systems

Low pressure drop

High efficient special filter media

Flow rate 10-170 l/min
Special sizes on request

Offline Filtration Systems

15-170 NFF2....
Operating temperature - 10°C to + 100°C

Application

Filtration of pressurised liquids and lubricants. Separate installation in the offline or cooling circuit for fine filtration and relief for main filter. Filtration of fresh oil and flushing of contaminated systems.
Wear protection of components and systems.

Design

Mobile filtration system mounted on 2- or 4-wheel-cart with EPE-series filter.

Material : as per spare parts list in this brochure.

Filter Element

Pleated design with optimised pleat density and various filter media. The filter elements is the most important component of the filter in view of prolonged life and wear protection of the system.
Oil cleanliness, the initial pressure drop and the dirt holding capacity are the most important criteria for selection.
For further detailed information please refer our "Filter Elements" brochure.

Accessories

Maintenance Indicators

For monitoring the filter element's contamination status, visual and visual/electrical indicators, with one or two switching points are available.

Bypass Valve

To protect the filter element during start up and over pressurisation due to clogging.

Vent Valve

For removing the air from the filter during starting and for safe de-pressurisation.

Particle Counter

Most industry today need to know the oil cleanliness level. We can provide an integrated particle counter to give results in ISO/SAEClasses. Software for connecting the computer to the particle counter is also possible.

Guideline for selection

Initial pressure drop : 0,1-0,5 bar at operating viscosity
Selection in accordance to flow and reservoir capacity

Reservoir capacity volume V	Circulation time time t	Circulation factor factor f*
< 1000 l	30 min	2
1000 l - 5000 l	60 min	1
5000 l - 10000 l	120 min	0,5
> 10000 l	180 min	0,33

Achievable oil cleanliness as per ISO 4406

Calculation example	H1XL	H3XL	H10XL
5	12/9/5	13/11/8	17/15/12
10	10/7/2	12/10/6	15/14/10

Reservoir capacity : *3500 l*
Recommended circulation factor : *1*
Needed pump flow:

$$Q = \frac{V \times f}{t} = \frac{3500 \times 1}{60} = 58{,}3 \text{ l/min}$$

Chosen size *: 80 l/min, 80 NFF2 0120*

** circulation factor (f) indicates how often the system capacity in liters (l) passes the filter system in one hour. The higher this factor the faster the flushing or cleaning process will be.*

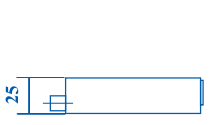
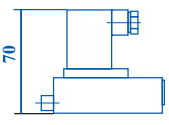
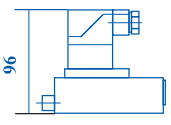
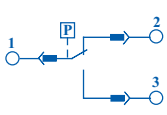
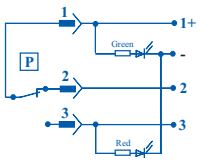
Filtering Media

The Main Filters are supplied with a variety of Elements including
Disposable Glass Fibre Media in 1, 3, 6, 10 & 20 µ(absolute)
Reusable SS Mesh in 10, 25, 40, 60 µ(nominal) & above
Disposable Paper Media in 5,10 & 25 µ(nominal)
Disposable Polypropylene Media in 10,25 & 40 µ(nominal)
Disposable Water Absorbing Media in 1, 3, 6, 10 & 20 µ(absolute)

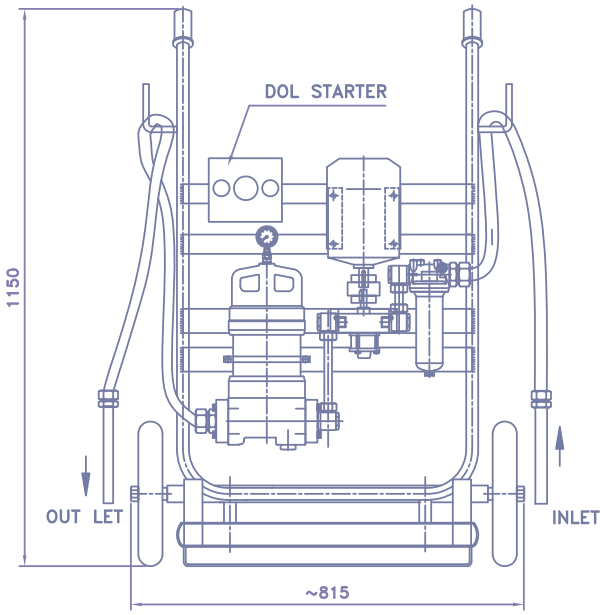
The Suction Filters are supplied with reusable SS Mesh Element of 800 µ(nominal) rating.

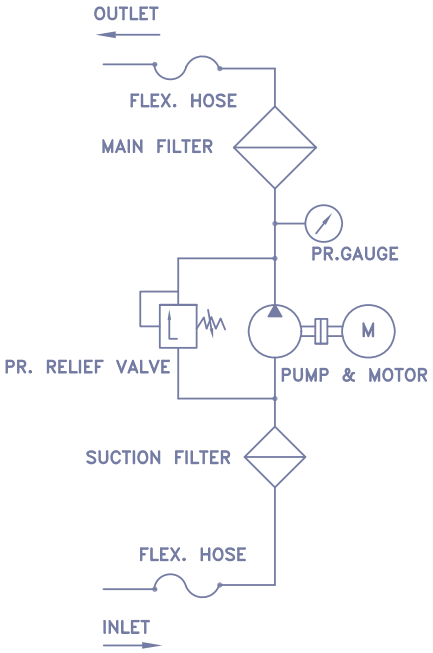
Maintenance Indicators

Maintenance indicators are used for monitoring the filter element's contamination status. They are available as visual or visual / electrical indicators.

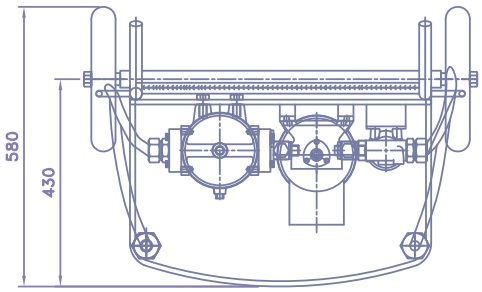
		
A... Visual	*B... Visual/electrical*	*C... Visual/electrical With two signal lights and two switching points*
Ordering information A2.5 = F2.5-A0-00-00-P A5.0 = F5.0-A0-00-00-P	*Ordering information B2.5 = F2.5-GW-02-00-P B5.0 = F5.0-GW-02-00-P*	*Ordering information C2.5 = F2.5-GW-26-00-P C5.0 = F5.0-GW-26-00-P*
	Switch Symbol 	*Switch Symbol* 

15-NFF2-0020.... 20-NFF2-0020.... 40-NFF2-0045
Dimensions





HYDRAULIC CIRCUIT

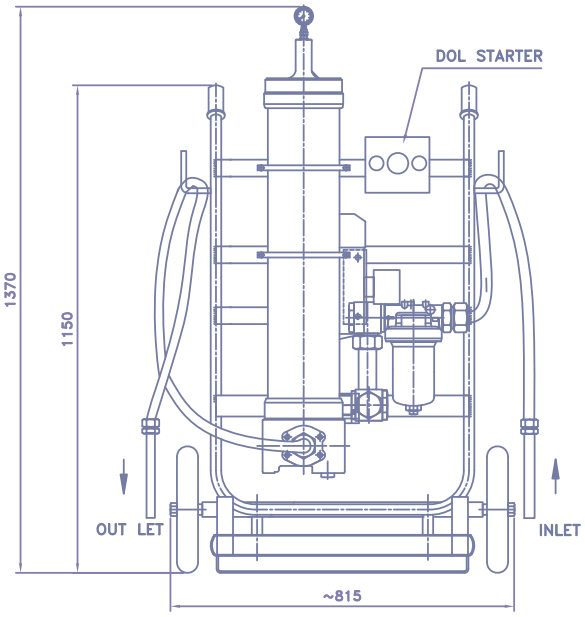


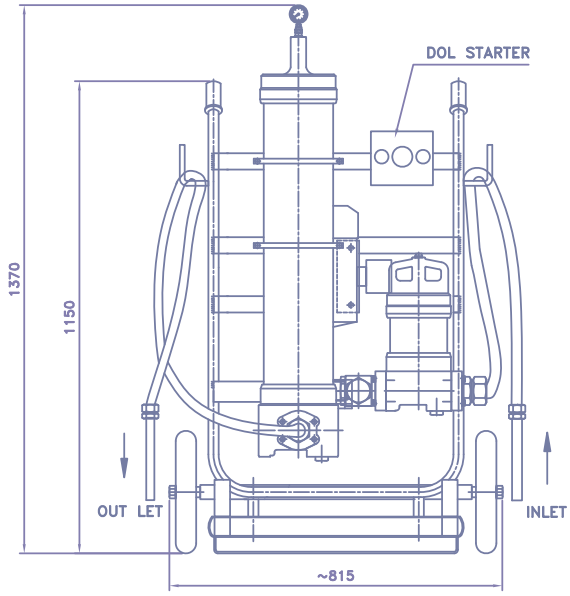
Ordering code / Spare Parts List

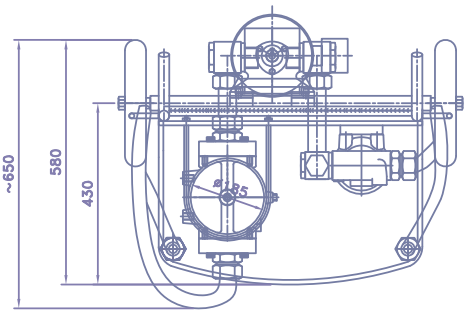
Filter System consisting of	*15 NFF2 0020...-A00-07A2, 5-00P00*	*20 NFF2 0020...-A00-07A2, 5-00P00*	*40 NFF2 0045...-A00-07A2, 5-00P00*
Volume Flow Rate	*15 l/min*	*20 l/min*	*40 l/min*
Main Filter	*40 FLE 0020...-A00-07A2, 5-R0P00*	*40 FLE 0020...-A00-07A2, 5-R0P00*	*40 FLE 0045...-A00-07A2, 5-R0P00*
Maintenance Indicator	*Visual (Pop-up / Gauge type), Electric Switch {F2,5 GW 02 00P}*		
Main Filter Element	*1.0020...-A00-0-P*	*1.0020...-A00-0-P*	*1.0045...-A00-0-P*
Suction Filter	*40 LE 0008 G800-A00-000-00P00*	*40 LE 0008 G800-A00-000-00P00*	*40 LE 0015 G800-A00-000-00P00*
Suction Filter Element	*2.0008 G800-A00-O-P*	*2.0008 G800-A00-O-P*	*2.0015 G800-A00-O-P*
Mounting	*Mobile with 2 wheels*		
Hose	*2m + lance 1m*		
Nominal Size Suction/Pressure	*Ø 16 / Ø 16*	*Ø 16 / Ø 16*	*Ø 25 / Ø 30*
Electrical Data	*415 V; 50 Hz; 1HP*	*415 V; 50 Hz; 1HP*	*415 V; 50 Hz; 1.5HP*
Pump Pressure	*Max 8 bar*		
Viscosity Range	*6-220 mm^2/s (up to 1000mm^2/s on request)*		

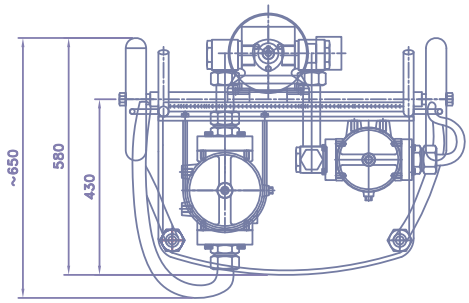
Filter fineness see Guideline for selection page 2, category "Filtering Media"
Special design on request

80-NFF2-0120.... 120-NFF2-0200
Dimensions







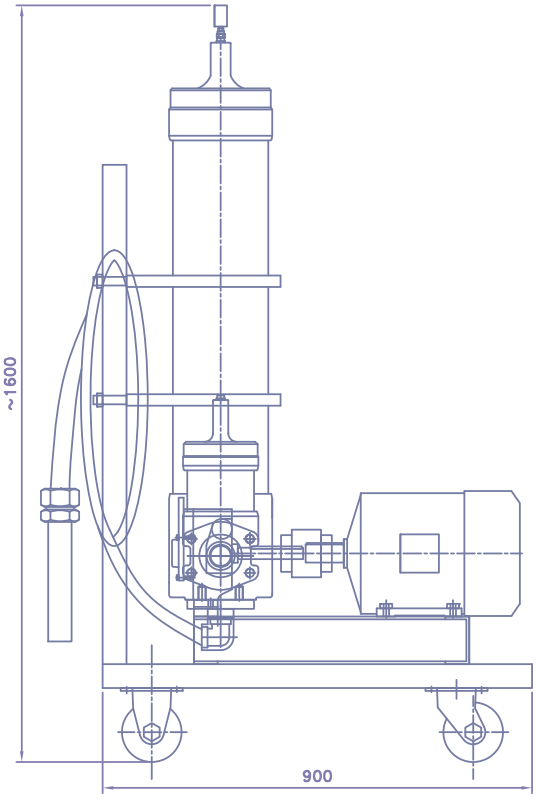


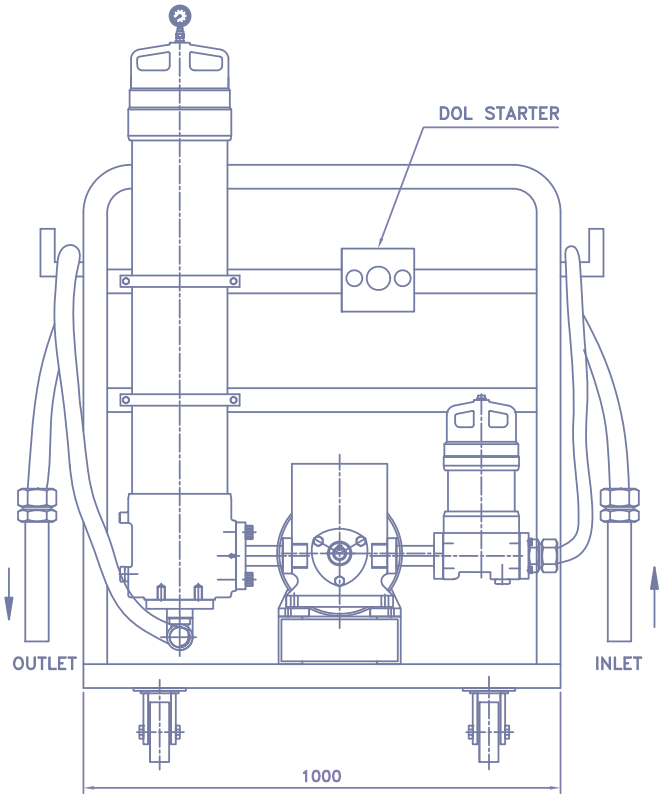
Ordering code / Spare Parts List

Filter System consisting of	*80 NFF2 0120...-A00-07A2, 5-00P00*	*120 NFF2 0200...-A00-07A2, 5-00P00*
Volume Flow Rate	*80 l/min*	*120 l/min*
Main Filter	*40 FLE 0120...-A00-07A2, 5-R0P00*	*40 FLE 0200...-A00-07A2, 5-R0P00*
Maintenance Indicator	*Visual (Pop-up / Gauge type), Electric Switch {F2,5 GW 02 00P}*	
Main Filter Element	*1.0120...-A00-0-P*	*1.0200...-A00-0-P*
Suction Filter	*40 LE 0020 G800-A00-000-00P00*	*40 FLE 0020 G800-A00-000-00P00*
Suction Filter Element	*2.0020 G800-A00-O-P*	*1.0020 G800-A00-O-P*
Mounting	*Mobile with 2 wheels*	
Hose	*2m + lance 1m*	
Nominal Size Suction/Pressure	*Ø 30 / Ø 45*	*1-1/2" / 1-1/2"*
Electrical Data	*415 V; 50 Hz; 3HP*	*415 V; 50 Hz; 5HP*
Pump Pressure	*Max 8 bar*	
Viscosity Range	*6-220 mm^2/s (up to 1000mm^2/s on request)*	

Filter fineness see Guideline for selection page 2, category "Filtering Media"
Special design on request

145-NFF2-0270.... 170-NFF2-0270
Dimensions





Ordering code / Spare Parts List

Filter System consisting of	*145 NFF2 0270...-A00-07A2, 5-00P00*	*170 NFF2 0270...-A00-07A2, 5-00P00*
Volume Flow Rate	*145 l/min*	*170 l/min*
Main Filter	*40 FLE 0270...-A00-07A2, 5-R0P00*	*40 FLE 0270...-A00-07A2, 5-R0P00*
Maintenance Indicator	*Visual (Pop-up / Gauge type), Electric Switch {F2,5 GW 02 00P}*	
Main Filter Element	*1.0270...-A00-0-P*	*1.0270...-A00-0-P*
Suction Filter	*40 FLE 0020 G800-A00-000-00P00*	*40 FLE 0020 G800-A00-000-00P00*
Suction Filter Element	*1.0020 G800-A00-O-P*	*1.0020 G800-A00-O-P*
Mounting	*Mobile with 4 wheels*	
Hose	*2m + lance 1m*	
Nominal Size Suction/Pressure	*2" / 2"*	*2" / 2"*
Electrical Data	*415 V; 50 Hz; 7.5HP*	*415 V; 50 Hz; 7.5HP*
Pump Pressure	*Max 8 bar*	
Viscosity Range	*6-220 mm^2/s (up to 1000mm^2/s on request)*	

Filter fineness see Guideline for selection page 2, category "Filtering Media"
Special design on request

Disposal Guidelines - Filters

Disposal

Before the filter is sent for disposal or recycling, it should always be de-pressurised completely. It is suggested that the filter is dismantled and the components disposed of as industrial waste.

Fluid residues are to be drained completely before disposal / recycle of the accumulator.

Filter Elements - Oil from the used filter elements is to be drained before the element is sent for disposal or recycling.

Decontaminate if needed and in accordance with local regulations.

Environmental Protection

Careless disposal of the product and/or residual fluid contained therein can cause environmental pollution.

Dispose the product in accordance with provisions applicable in the country of use.

Fluid residues are to be disposed according to the respective safety data sheets (MSDS) valid for the specific hydraulic fluids.

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Internet : www.epe-india.com
E-mail : business@epe-india.com

Catalogue Ref. : EPE/Disp/15-16/002

Disposal Guidelines - Accumulators

Disposal

Before the accumulator is sent for disposal or recycling, it should always be discharged completely of the pre-charge pressure and the gas valve unscrewed. Pre-loading and checking kits are suited for this task.

For permanently sealed design Diaphragm Accumulators (Version-P) the diaphragm is preferably damaged on a test rig or the accumulator is carefully drilled (Ø ≥ 6 mm) on the gas chamber using a suitable clamp. As the gas flowing out can draw metal splinters or particles with it, safety glasses must be worn.

Fluid residues are to be drained completely before disposal / recycle of the accumulator. Decontaminate if needed and in accordance with local regulations.

Environmental Protection

Careless disposal of the product and/or residual fluid contained therein can cause environmental pollution.

Dispose the product in accordance with provisions applicable in the country of use.

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